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ORIGINAL PAPER

Lightweight Vision Transformer with CBAM and LSTM for Efficient Violence Detection in Image Sequences

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Abstract

In recent years, the need for efficient violence detection methods has become increasingly critical for various security applications. Existing approaches leveraging deep learning (DL), particularly Vision Transformers (ViTs), often suffer from high complexity and overhead, limiting their real-time applicability. This paper proposes a lightweight spatiotemporal transformer-based violence detection method that integrates ViT with a Convolutional Block Attention Module (CBAM) and Long Short-Term Memory (LSTM) for spatial and temporal feature extraction, respectively. By replacing the multi-head self-attention (MSA) mechanism in ViT with CBAM, we significantly reduce the number of parameters and computational costs while enhancing the model's ability to focus on localized patterns essential for violence detection. Our approach is validated on four benchmark datasets, achieving accuracies of 99.81%, 99%, 99.50%, and 99.10% on AIRTLab, Industrial Surveillance, RWF2000, and Hockey Fights, respectively. The proposed method demonstrates a good balance between model complexity and computational efficiency, making it suitable for real-time violence detection applications.

Keywords CBAM · Deep learning · LSTM · Violence detection · ViT

1 Introduction

Violence detection (VD) aims to identify any violent activity performed by humans in public places through image and video sequences [1]. However, automatic violence detection remains challenging due to irregular motion, occlusions, varying viewpoints, and low resolution. To address these issues, many machine and DL methods have been explored, out of which convolutional neural networks (CNNs) and their variants have been primarily exploited for their excellent performance in the extraction of local spatial features [2, 3]. It has often been combined with LSTM, bidirectional LSTM (BiLSTM), and convolutional LSTM (ConvLSTM) for the extraction of spatiotemporal features [4–7]. Additionally, transfer learning (TL)-based approaches have been proposed using different variants of CNN2D and CNN3D, such as MobileNet, VGG16, and VGG19, and a few other combinations, such as federated learning (FL) and LSTM, have also been introduced [8–10]. The existing methods are more inclined towards the attention modules and transformers due to their ability to extract global and local dependencies within feature maps [11–13]. Attention modules such as spatial attention module (SAM), channel attention module (CAM), CBAM and self-attention (SA) have been utilized in the classification process to extract meaningful features [14, 15]. It is to be noted that a few research works have integrated transformers-based MSA modules with CNN-LSTM architecture for violence detection [16, 17]. ViT and shifted window (SWIN) transformers utilized SA modules to extract long-term and global dependencies between image patches [18]. However, transformer-based SA mechanisms can lead to high complexity, overhead and significant computational costs. Since violent activity is often spontaneous and hasty, there is a great need to focus more on localized patterns within image frames rather than global features.

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To mitigate the limitations of the existing transformer-based SA methods, we propose a modified architecture of ViT that employs the CBAM module in place of the MSA mechanism. CBAM improves the model's ability to concentrate on the most significant spatial and channel-wise features, which are crucial for identifying localized patterns to classify violent activities and also reduces computational complexity parameters counts, results in improved feature extraction and a lightweight model. Additionally, the proposed method incorporates an LSTM layer to learn the long-range dependencies between the image patches.

The key contributions of the proposed method are as follows:

- We propose a lightweight spatiotemporal framework that integrates ViT, CBAM and LSTM for the classification of violent activities.
- The lightweight architecture has been implemented by replacing the standard MSA in ViT with CBAM, which significantly reduced the complexity, overhead and computational cost with a reduced number of parameters.
- The integration of CBAM and LSTM enables the proposed method to learn significant features by focusing on localized and global patterns.
- The proposed methodology is evaluated with four publicly available benchmark datasets, AIRTLab, Industrial Surveillance (IS), RWF2000, and Hockey Fights, which attained an accuracy of 99.81%, 99%, 99.50%, and 99.10%, respectively, indicating its robustness and efficiency for real-time applications.

2 Related works

2.1 Deep learning-based methods

A violence detection method called spatial-temporal action translation (STAT) [19] employed generative adversarial networks (GANs) to construct images based on extracted motion features. Ullah et al. [20] proposed a violence detection network (VDNet) based on an industrial Internet of Things (IIOT) framework. This work utilizes CNN2D for suspicious human activity and ConvLSTM and gated recurrent unit (GRU) for classifying violent activities. Expanding on spatiotemporal features, a study [21] presented three hybrid architectures – 3DCNN with support vector machines (SVM), fully connected layers and ConvLSTM. Further advancing spatiotemporal representation, Hwang and Kang (2023) [22] introduced a violence detection model that detects violent activity with merged image frames. These merged frames were processed through CNN3D, ResNet50 and CBAM. Futhermore, TL techniques, such as VGG16-ConvLSTM and VGG19-ConvLSTM, ResNet50, Inception-Resnet152V2, and DenseNet121 have also been introduced and found helpful in detecting violent activities [23, 24].

2.2 Attention and transformers-based methods

Mahmoodi and Nezamabadi-pour [25] proposed an attention-based model to exploit 2DCNN and squeeze temporal attention (STA) module to extract spatiotemporal features. This framework was integrated with the entropy spatial module (ESM) and SAM for improved performance. Another approach is presented by Deshpande et al. [26] that employed K-Means clustering for extracting keyframes and a residual attention framework, ResAttenConvLSTM2D. Similarly, Chaturvedi et al. [27] introduced a hybrid approach that utilized ResNet50 and spatial and channel wise attention-based ConvLSTM (SCAN-ConvLSTM) encoder for feature extraction.

Singh et al. [28] proposed a framework that utilized ViT to extract features from augmented image frames. Based on the fusion strategy of local and global features, Zhang and Tian [29] proposed a framework based on SWIN transformer and attention module to extract global and local dependencies for anomaly classification from image sequences. For pose detection in violence-based videos, Zhou [30] presented an end-to-end framework that employed TokenPose-based ViT for pose estimation and CNN3D for spatiotemporal feature extraction.To enhance this approach, Zhou [31] incorporated ResNet50 and replaced the existing transformer architecture with object (ODT) and keypoint detection transformers (KDT). Another transformer-based three-stage framework was introduced by Lee and Kang [32] that used pretrained ViT, BiLSTM and MSA modules, followed by a transformer encoder block for extracting spatiotemporal features. To enhance the existing violence detection methods, a lightweight VDNet that employed Special Task Temporal Convolution Networks (ST-TCN) blocks, bottleneck attention, and transformer blocks for feature extraction from collected Wi-Fi and visual sensor-based dataset was proposed. [33].

From the existing literature review, we have found that the existing deep learning approaches with CNN, including its pretrained variants, have demonstrated comparable performance with reasonable accuracy. It has also been seen that approaches utilizing the ViT architecture have attained reasonable accuracy, but high overhead associated with MSA modules and increased number of computational parameters. To address this issue, in this work, we propose a transformer-based framework that focuses on reducing the computational overhead and parameters of ViT by incorporating a CBAM attention module to enhance the classification of violent activities.

3 Proposed method

3.1 Proposed VD method

It was found that the datasets have some video frames that are blank or blurred due to the spontaneous nature of violent activity. To mitigate this issue, the elimination of the blank frames is performed in the preprocessing phase of the proposed methodology. In the proposed framework, each image frame is resized to 192 x 192 x 3 and partitioned into 144 non-overlapping patches of size 16 x 16 x 3. These patches are passed to a convolutional layer, which maps each patch to a 768-dimensional embedding, which results in an output tensor of shape 12 x 12 x 768, as illustrated in Fig. 1(a). These patches are then passed to the positional embedding block and then processed through modified encoder block. The resultant spatial features are then passed to an LSTM layer with 1024 units for temporal feature extraction. These spatiotemporal features are passed through four fully connected layers with 512, 128, 16 units and a leaky ReLU activation function, followed by the final layer with two units and a sigmoid activation function for binary classification of violent activities. To minimize overfitting of the model, the dropout and batch normalization layers are integrated into the model.

In the proposed method, LSTM layer offers us three significant advantages. First, in the context of local features, each processed patch consists of significant localized information. By treating these patches as a sequence, the model is able to find crucial cues that signify violent behavior. Second, the temporal awareness of LSTM allows it to maintain a memory of previous inputs, enabling the model to interpret relationships between patches and extract global dependencies. This capability is beneficial for understanding the spatial arrangements of features when identifying violent activities. Third, processing the patches in place of a sequence of image frames enhances efficiency by reducing computational costs and load and allowing the model to focus on the relevant features in classifying violent activities.

3.2 Modified transformer encoder block

In the proposed methodology, the base ViT is exploited, featuring an embedded dimension of 768 and 12 stacked encoder blocks. As discussed previously, SA plays a pivotal role in ViT, which is responsible for extracting global dependencies between feature maps shown in Fig. 1(b), which results in high complexity and overhead in the architecture. To solve this problem, we introduce a modified transformer encoder block that utilizes CBAM in place of MSA, which yields a lightweight and less overhead architecture and focuses on extracting relevant local patterns present in the patches by incorporating spatial and channel-wise attention mechanisms. In the modified framework, the flattening of patches is skipped. These embedded patches with dimensions 12 x 12 x 768 are reshaped to match the input shape for CBAM architecture, that is, (batchsize, 12, 12, 3) to (batchsize, 3, 12, 12). Regarding the feedforward network (Multi-Layer Perceptron (MLP)), a reshape layer is added before and after the MLP layer to convert the output feature map from 2D to 1D and vice versa to perform residual operations shown in Fig.1(c). This operation from CBAM to the MLP layer is repeated 11 times, corresponding to the 12 encoder layers for the base ViT model. The resultant output is then fed to the normalization layer. To evaluate the effect of SA on the proposed methodology, another framework is designed where an SA module is appended after the CBAM in the proposed framework, depicted in Fig. 1(d).

4 Experimental analysis

This section provides a detailed discussion of the proposed method over four publicly available benchmark datasets: AIRTLab, IS, RWF2000, and Hockey Fights. The performance analysis of the proposed method has been represented in terms of confusion matrix, accuracy, training/validation accuracy and loss per epoch curves. The comparative analysis is performed to evaluate the performance of the proposed method based on accuracy. The implementations are conducted in the Google Colaboratory Pro environment with the Pytorch library. The proposed framework exploits SGD optimizer and categorical cross-entropy loss function for each dataset. In each dataset, the split ratio (train: validation: test) is taken as 80:15:5.

4.1 Datasets

This section briefly details the publicly available benchmark datasets utilized: AIRTLab [34], IS [20], RWF2000 [35], and Hockey Fights [36] in Table 1. The hyperparameters utilized for the implementations are presented in Table 2.

4.2 Discussions

4.2.1 Performance Analysis

To evaluate the efficiency of the proposed model, we conducted experiments analyzing the impact of incorporating the SA module. The performance of the proposed models with and without SA is presented in Table 3 based on accuracy.

The proposed ViT-CBAM + LSTM demonstrates better performance with attained accuracies of 99.81%, 99%, 99.50%, and 99.10% on AIRTLab, IS, RWF2000, and Hockey Fights, respectively. In contrast, the ViT-CBAM+

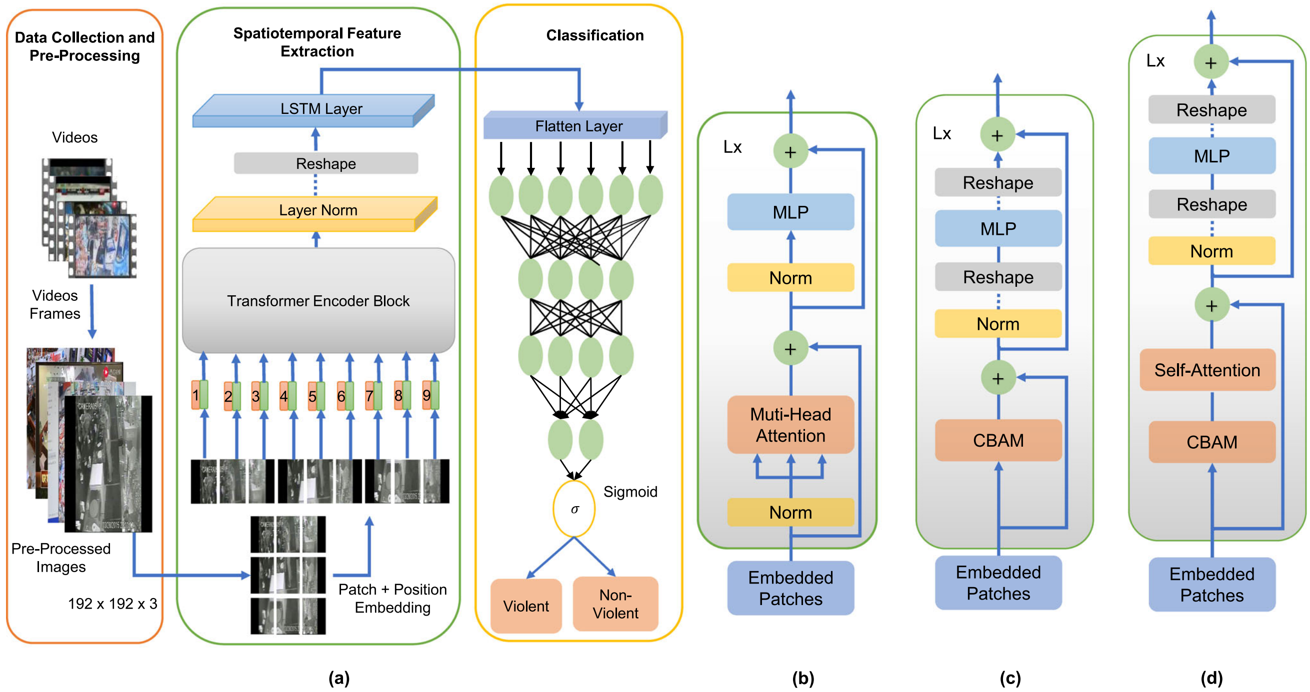

Fig. 1 Architecture of the (a) Proposed Method (b) Traditional Multi-Head Self Attention Module (c) Designed Transformer Encoder Block incorporating CBAM module (d) Designed Transformer Encoder Block integrating CBAM and SA modules

Table 1 Dataset details

Parameters/Datasets	AIRTLab	IS	RWF2000	Hockey Fights
Total Videos	350	300	2000	1000
Number of Violent Videos	230	150	1000	500
Number of Non-Violent Videos	120	150	1000	500
Dimensions	$1920\times1080\times3$	Variable	Variable	$360\times288\times3$
FPS	30	20–30	30	30
Duration of videos	2–3 sec	5 sec	5 sec	1–2 sec

Table 2 Hyperparameters utilized for the proposed framework

Hyperparameters / Datasets	AIRTLab	IS	RWF2000	Hockey Fights
Number of frames taken	30	40	20	20
Number of videos considered	350	300	1000	1000
Total number of images	10500	12000	20000	20000
Batch Size	75	50	100	100
Momentum	0.9	0.9	0.9	0.9
Epochs	25	25	25	25

Table 3 Accuracy (%) for proposed framework, ViT-CBAM + LSTM

Datasets / Configuration	AIRTLab	IS	RWF2000	Hockey Fights
Without SA	99.81	99.00	99.50	99.10
With SA	99.42	97.83	99.10	97.00

Table 4 Comparison analysis of the proposed framework and pretrained (P) models based on the number of parameters and accuracy (%)

Model / Datasets	AIRTLab	IS	RWF2000	Hockey Fights	Parameters (Only ViT/SWIN)
ViT_Base_16 (P)	98.28	99.16	99.50	99.70	86M
SWIN_B (P)	98.71	93.17	83.30	96.40	88M
ViT_CBAM_LSTM	99.81	99.00	99.50	99.10	**58M**
ViT_CBAM_SA_LSTM	99.42	97.83	99.10	97.00	67M

SA + LSTM model attained lower accuracies of 99.42%, 97.83%, 99.10%, and 97% on AIRTLab, IS, RWF2000, and Hockey Fights, respectively, which indicates that the introduction of SA leads to increased complexity and overhead, resulting in degradation of performance.

To further validate the performance of the proposed methodology, a comparative analysis is conducted against the pretrained transformer (ViT and SWIN) models, presented in Table 4. The results indicate that the proposed ViT-CBAM + LSTM model outperforms in the case of AIRTLab (99.81%) and RWF2000 (99.50%) datasets. In contrast,

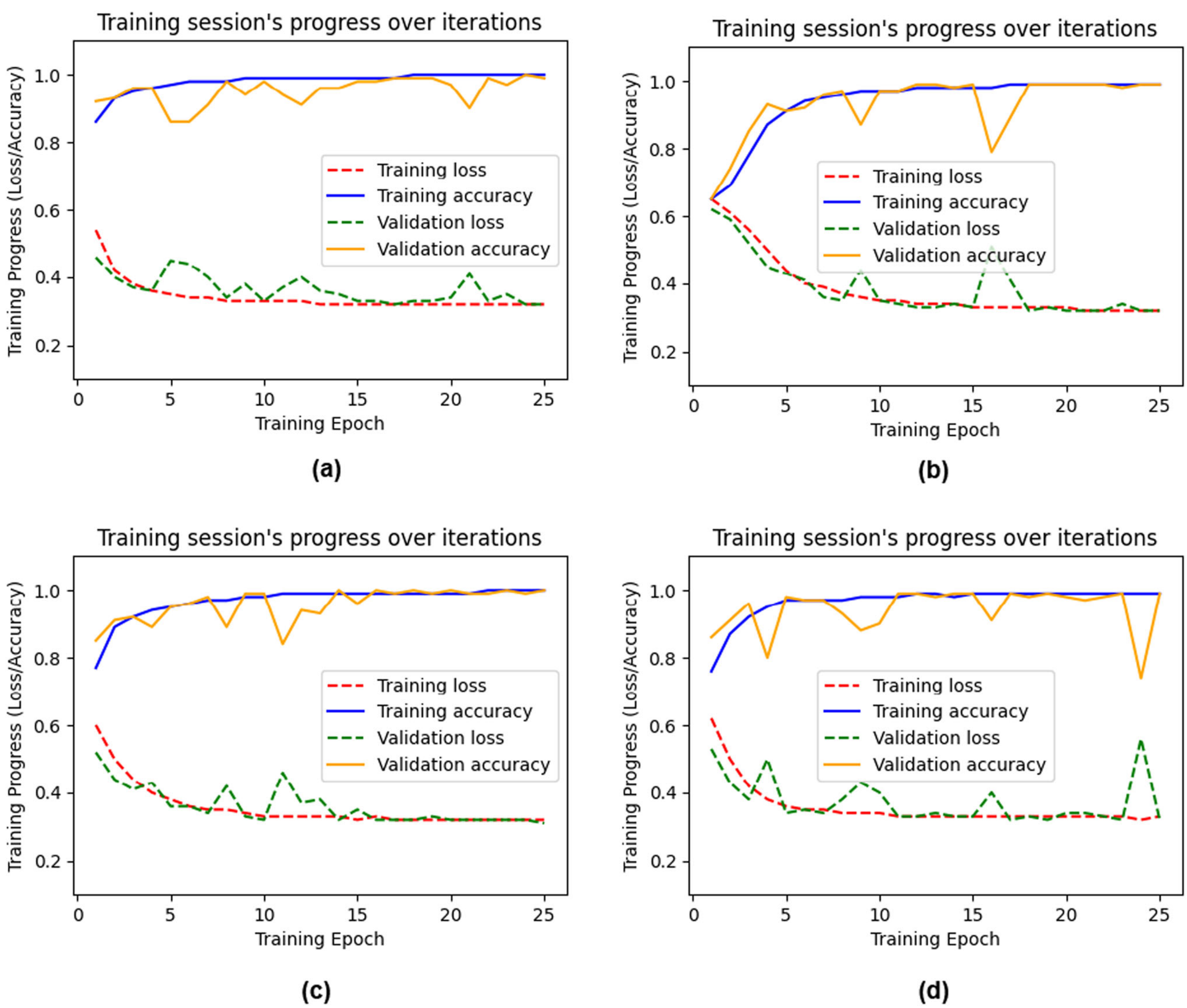

Fig. 2 Training/Validation Accuracy and Loss per Epoch graph for (a) Hockey Fights, (b) RWF2000, (c) AIRTLab, (d) Industrial Surveillance datasets using ViT-CBAM-LSTM

Table 5 Confusion Matrices for (a) Hockey Fights, (b) RWF2000, (c) AIRTLab, (d) Industrial Surveillance datasets using ViT-CBAM-LSTM

Dataset	Hockey Fights		RWF2000		AIRTLab		IS	
Activity	V	NV	V	NV	V	NV	V	NV
V	466	7	490	4	346	0	280	0
NV	2	525	1	505	1	178	6	314

V: violent; NV: non-violent

the pretrained ViT model achieves higher accuracy on IS (99.16%) and Hockey Fights (99.7%) datasets. Regarding only ViT architecture, the proposed ViT with CBAM has only 58M parameters, significantly lower than those of pretrained ViT, SWIN, and ViT-CBAM with SA having 86M, 88M, and 67M, respectively. This reduction also contributed to lower computational costs, less overhead and faster training time.

In the case of the Hockey Fights dataset, minor spikes are observed at some intervals, illustrated in Fig. 2(a), suggesting slight overfitting. Correspondingly, in the confusion matrix in Table 5, the model misclassified seven non-violent and two violent activities. In the case of the RWF2000 dataset, the proposed method encountered sharp spikes between epochs 5-10 and 15-17, depicted in Fig. 2(b). However, the training curve stabilizes and converges after the 17th epoch, suggesting the absence of overfitting. The confusion matrix reveals better performance of the proposed method, with one violent and four non-violent cases misclassified.

In the case of the AIRTLab dataset, it has been observed that there are a few spikes between 5 and 15 epochs; after the 15th epoch, the curves converge, show in Fig. 2(c), which indicates the absence of overfitting. The confusion matrix confirmed this observation with only one violent misclassified and perfect classification of all non-violent activities. In the case of the IS dataset, the proposed model exhibited notable fluctuations throughout the training process, suggesting mild overfitting that indicates the presence of an overfitting case, resulting from the dataset having vast backgrounds, where resizing might lead to loss of contextual data, illustrated in Fig 2(d). This observation is confirmed by the confusion matrix, where the proposed model failed to recognize six violent activities but detected all non-violent activities.

4.2.2 Comparative Analysis

This section presents a comparative analysis of the proposed ViT-CBAM + LSTM (with and without SA) with DL and transformer-based approaches based on attained accuracy.

To evaluate the performance of Hockey Fights, we compare with DL and TL algorithms employing CNN architecture viz., [19, 20, 33] and [23]. We have also compared with attention-based methods such as [16, 26, 37, 38] and [27]. Some transformer-based methods employed ViT such as [28, 39] and [32], presented in Table 6. The table shows that the proposed methodology outperformed all methods except SCAN-ConvLSTM [27].

To evaluate the performance of the RWF2000 dataset, we compare with DL and TL algorithms employing CNN algorithm such as, [19, 20] and [24]. We have also compared with attention-based methods viz., [22, 37, 38], and [27]. Some transformer-based methods, such as, [30, 39] and [31] summarized in Table 7 are compared with the proposed method. From the table, we can say that the proposed method outperformed existing works.

To evaluate the performance of IS, we compare with DL algorithms employing CNN and transformers architecture, such as, [20] and [40]. The comparative analysis is presented in Table 8. From the table, we can say that the proposed methodology outperformed existing methods.

To evaluate the performance of the AIRTLab dataset, we compare the proposed method with DL and TL algorithms viz., [19, 21, 41] and [23]. We have also compared attention-based and transformer-based methods viz., [26] and [40], summarized in Table 9. The table shows that the proposed methodology outperformed all methods.

The experimental analysis shows the effectiveness of the proposed model, summarized as follows.

- The proposed ViT-CBAM architecture has fewer parameters and training time than the standard ViT framework, which is 58M. The proposed methodology outperforms all transformer-based methods by 2-5%.
- The RWF2000 dataset contains videos with low resolutions and irregular content, attained 99.50% accuracy, indicating the better performance of the proposed model.
- It has been observed from the Figs. 2(a) and 2(d) that there are spikes in the training/validation accuracy and loss for Hockey Fights and IS datasets, indicating overfitting from irregularity and vast backgrounds in image frames.
- The proposed method with SA showed a decline in accuracy due to misclassification caused by increased complexity and overhead, depicted in Table 3.

5 Conclusions

In this work, we have proposed a lightweight spatiotemporal transformer-based violence detection model that employed ViT and LSTM for spatial and temporal feature extraction, respectively, followed by fully connected layers for binary classification. Although ViT architecture provides better results, it delivers high overhead due to SA and multiple heads, leading to increased computational parameters, complexity and limited efficiency. Focusing on this short-

Table 6 Comparative Analysis of the Hockey Fights Dataset

Year	Ref	Approach	Accuracy (%)
2021	[16]	MSA + Bi-ConvLSTM	97.50
2021	[37]	SAM + GhostNet + ConvLSTM	97.50
2021	[20]	ConvLSTM + GRU	98.50
2022	[28]	ViViT	97.14
2022	[38]	Fusion CNN + SAM + Bi-ConvLSTM (temporal attention)	89.10
2023	[26]	ResAttenConvLSTM2D	96.00
2023	[39]	ViT-based mixture of experts (MoE)	98.20
2024	[23]	VGG16 + ConvLSTM	94.00
		VGG19 + ConvLSTM	94.00
2024	[19]	STAT	96.00
2024	[33]	VDNet	98.50
2024	[32]	CNN2D + Bi-LSTM + ViT	98.00
2024	[27]	SCAN-ConvLSTM	99.83
–	–	**ViT-CBAM-LSTM**	**99.10**
–	–	**ViT-CBAM-SA-LSTM**	**97.00**

Table 7 Comparative Analysis of the RWF2000 Dataset

Year	Ref	Approach	Accuracy (%)
2021	[20]	ConvLSTM + GRU	88.20
2021	[37]	SAM + GhostNet + ConvLSTM	87.50
2022	[30]	ViT + 3DCNN	89.00
2022	[38]	Fusion CNN + SAM + Bi-ConvLSTM (temporal attention)	89.10
2023	[22]	CBAM + ResNet10 (Conv3D)	99.20
2023	[31]	ODT + KDT + CNN3D	83.20
2023	[39]	ViT-based MoE	92.40
2024	[19]	STAT	98.75
2024	[27]	SCAN-ConvLSTM	84.32
2025	[24]	Ensemble Networks + LSTM	92.70
–	–	**ViT-CBAM-LSTM**	**99.50**
–	–	**ViT-CBAM-SA-LSTM**	**99.10**

Table 8 Comparative Analysis of the IS Dataset

Year	Ref	Approach	Accuracy (%)
2021	[20]	ConvLSTM + GRU	80.00
2024	[40]	SAM + SWIN-B	98.85
–	–	**ViT-CBAM-LSTM**	**99.00**
–	–	**ViT-CBAM-SA-LSTM**	**97.83**

coming, we exploited a CBAM module placed in place of MSA, which reduced the number of parameters from 86M to 58M when considering only ViT architecture. This work also highlighted the effect of SA in the proposed methodology, which increased the number of parameters to 67M and training time. The proposed framework is evaluated with four publicly available benchmark datasets, viz, AIRT-Lab, Hockey Fights, IS, and RWF 2000, which attained an

Table 9 Comparative Analysis of the AIRTLab Dataset

Year	Ref	Approach	Accuracy (%)
2021	[21]	CNN3D + SVM	96.10
		CNN3D + FC	95.62
		ConvLSTM	97.15
2022	[41]	CNN2D + ConvLSTM	98.29
2023	[26]	ResAttenConvLSTM2D	90.00
2024	[6]	CNN2D + ConvLSTM	90.47
2024	[23]	VGG16 + ConvLSTM	88.68
		VGG19 + ConvLSTM	88.68
2024	[19]	STAT	94.00
2024	[40]	SAM + SWIN-B	98.28
–	–	**ViT-CBAM-LSTM**	**99.81**
–	–	**ViT-CBAM-SA-LSTM**	**99.42**

accuracy of 99.81%, 99.10%, 99%, and 99.50%, respectively. The comparative analyses conclude that our proposed method attained better performance and reasonable accuracy, indicating a good balance between the model complexity and computational efficiency. The proposed work can be enhanced for temporal feature extraction and can be integrated with different attention modules for binary and multi-class violence detection.

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Data Availability The data associated with this work is available at https://github.com/Aishvarya112/ViT_CBAM-Violence-Detection.

Declarations

Conflict of interest The authors declare that there is no conflict of interest regarding this manuscript.

Competing interests The authors declare no competing interests.

References

1. Kaur, G., Singh, S.: Revisiting vision-based violence detection in videos: a critical analysis. Neurocomputing **597**, 128113 (2024)
2. Kang M.S., Park R.H., Park H.M.: Efficient spatio-temporal modeling methods for real-time violence recognition. IEEE Access **9**, 76270–76285 (2021)
3. Mahmoodi, J., Nezamabadi-pour, H., Abbasi-Moghadam, D.: Violence detection in videos using interest frame extraction and 3D convolutional neural network. Multimed. Tools Appl. **81**(15), 20945–20961 (2022)
4. Asad, M., Yang, J., He, J., Shamsolmoali, P., He, X.: Multi-frame feature-fusion-based model for violence detection. Vis. Comput. **37**(6), 1415–1431 (2021)
5. Halder, R., Chatterjee, R.: CNN-BiLSTM model for violence detection in smart surveillance. SN Comput. Sci. **1**(4), 201 (2020)
6. Hlavata, R., Kamencay, P., Sykora, P., Hudec, R., Sturekova, J.: Recognition of abnormal behavior from a thermal camera using deep learning In: 2024 34th International Conference Radioelektronika (RADIOELEKTRONIKA), pp. 1–4. IEEE (2024)
7. Contardo, P., Tomassini, S., Falcionelli, N., Dragoni, A.F., Sernani, P.: Combining a mobile deep neural network and a recurrent layer for violence detection in videos In: CEUR Workshop Proceedings 3402, pp. 35–53. CEUR-WS (2023)
8. Kukarni, M., Chakraborty, R.: Violent activity detection on public transportation using surveillance footage In: 2023 16th International Conference on Developments in eSystems Engineering (DeSE), pp. 737–742. IEEE (2023)
9. Quentin, P., Swan, S., Hugo, W., Leo, R., Siba, H., Antoun, Y.: Balancing accuracy and training time in federated learning for violence detection in surveillance videos: a study of neural network architectures. arXiv:2308.05106 (2023)
10. Negre, P., Alonso, R.S., Prieto, J., García, O.: Video violence detection using pre-trained VGG19 combined with manual logic, LSTM layers and Bi-LSTM layers. Appl. Intell. **56**(3), 72 (2026)
11. Hameed, S., Amin, J., Anjum, M.A., Sharif, M.: Suspicious activities detection using spatial-temporal features based on vision transformer and recurrent neural networks. J. Ambient. Intell. Humaniz. Comput. **15**(9), 3379–3391 (2024)
12. Li, C., Yang, X., Liang, G.: Keyframe-guided video swin transformer with multi-path excitation for violence detection. Comput. J. **67**(5), 1826–1837 (2024)
13. Pareek, G., Nigam, S., Singh, R.: Modeling transformer architecture with attention layer for human activity recognition. Neural Comput. Appl. **36**(10), 5515–5528 (2024)
14. Abbass M.A.B., Kang H.S.: Violence detection enhancement by involving convolutional block attention module into various deep learning architectures: comprehensive case study for ubi-fights dataset. IEEE Access **11**, 37096–37107 (2023)
15. Shao, W., Rajapaksha, P., Wei, Y., Li, D., Crespi, N., Lou, Z.: COVAD: Content-oriented video anomaly detection using a self attention-based deep learning model. Virtual Reality & Intelligent Hardware **5**(1), 24–41 (2023)
16. Rendón-Segador, F.J., Álvarez-García, J.A., Enríquez, F., Deniz, O.: Violencenet:Dense multi-head self-attention with bidirectional convolutional lstm for detecting violence. Electronics **10**(13), 1601 (2021)
17. Vosta, S., Yow, K.C.: KianNet: A violence detection model using attention-based CNN-LSTM structure. IEEE Access **12**, 2198–2209 (2023)
18. Han, K., et al.: A survey on vision transformer. IEEE Trans. Pattern Anal. Mach. Intell. **45**(1), 87–110 (2022)
19. Ehsan, T.Z., Nahvi, M., Mohtavipour, S.M.: An accurate violence detection framework using unsupervised spatial-temporal action translation network. Vis. Comput. **40**(3), 1515–1535 (2024)
20. Ullah, F.U.M., Muhammad, K., Haq, I.U., Khan, N., Heidari, A.A., Baik, S.W., De Albuquerque, V.H.C. AI-assisted edge vision for violence detection in IoT-based industrial surveillance networks. IEEE Trans. Industr. Inf. **18**(8), 5359–5370 (2021)
21. Sernani, P., Falcionelli, N., Tomassini, S., Contardo, P., Dragoni, A.F.: Deep learning for automatic violence detection: tests on the AIRTLab dataset. IEEE Access **9**, 160580–160595 (2021)
22. Hwang, I.C., Kang, H.S.: Anomaly detection based on a 3d convolutional neural network combining convolutional block attention module using merged frames. Sensors **23**(23), 9616 (2023)
23. Haque, M., et al.: Generalising violence detection with a new near-real-world violence dataset. In: 2024 3rd International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE), pp. 1–6. IEEE (2024)
24. Kaur, G., Singh, S.: An ensemble based approach for violence detection in videos using deep transfer learning. Multimed. Tools Appl. **84**(12), 11001–11025 (2025)
25. Mahmoodi, J., Nezamabadi-pour, H.: A spatio-temporal model for violence detection based on spatial and temporal attention modules and 2DCNNs. Pattern Anal. Appl. **27**(2), 46 (2024)
26. Deshpande, A., Warhade, K., Sanap, P.: Abnormal activity recognition with residual attention-based ConvLSTM architecture for video surveillance. International Journal of Intelligent Engineering & Systems **16**(6) (2023)
27. Chaturvedi, K., Dhiman, C., Vishwakarma, D.K.: Fight detection with spatial and channel wise attention based ConvLSTM model. Expert. Syst. **41**(1), e13474 (2024)
28. Singh, S., Dewangan, S., Krishna, G.S. Tyagi, V., Reddy, S., Medi, P.R.: Video vision transformers for violence detection. arXiv:2209.03561 (2022)
29. Zhang, M., Tian, X.: Transformer architecture based on mutual attention for image-anomaly detection. Virtual Reality & Intelligent Hardware **5**(1), 57–67 (2023)

30. Zhou, L.: End-to-end video detection with transformer. In: 2022 5th International Conference on Pattern Recognition and Artificial Intelligence (PRAI), pp. 880–884. IEEE (2022)
31. Zhou, L., Li, W., Chen, Y., Liu, H., Yang, M., Liu, Z.: Human keypoint change detection for video violence detection based on cascade transformer. In: 2023 IEEE 6th International Conference on Pattern Recognition and Artificial Intelligence (PRAI), pp. 88–94. IEEE (2023)
32. Lee, J.W., Kang, H.S.: Three-stage deep learning framework for video surveillance. Appl. Sci. **14**(1), 408 (2024)
33. Khan, M., El Saddik, A., Gueaieb, W., De Masi, G., Karray, F.: VD-Net: An edge vision-based surveillance system for violence detection. IEEE Access **12**, 43796–43808 (2024)
34. Bianculli, M., Falcionelli, N., Sernani, P., Tomassini, S., Contardo, P., Lombardi, M., Dragoni, A.F.: A dataset for automatic violence detection in videos. Data in Brief **33**, 106587 (2020)
35. Cheng, M., Cai, K., Li, M.: RWF 2000: An open large scale video database for violence detection. In: 2020 25th International Conference on Pattern Recognition (ICPR), pp. 4183–4190. IEEE (2021)
36. Bermejo Nienas, E., Deniz Suarez, O., Bueno Gracía, G., Sukthankar, R.: Violence detection in video using computer vision techniques. In: International Conference on Computer Analysis of Images and Patterns, pp. 332–339. Springer (2011)
37. Liang, Q., Li, Y., Yang, K., Wang, X., Li, Z.: Long-term recurrent convolutional network violent behaviour recognition with attention mechanism. In: MATEC Web of Conferences, **336**, p. 05013. EDP Sciences (2021)
38. Asad, M., Jiang, H., Yang, J., Tu, E., Malik, A.A.: Multi-level two-stream fusion-based spatio-temporal attention model for violence detection and localization. Int. J. Pattern Recognit Artif Intell. **36**(01), 2255002 (2022)
39. Mohammadi, H., Nazerfard, E., Firoozi, T.: Reinforcement learning based mixture of vision transformer for video violence recognition. arXiv:2310.03108 (2023)
40. Garg, A., Nigam, S., Singh, R., Shastri, A., Singh, M.: Spatial attention transformer based framework for anomaly classification in image sequences. In: International Conference on Intelligent Human Computer Interaction, pp. 53–64. Springer (2023)
41. Vrskova, R., Hudec, R., Kamencay, P., Sykora, P.: A new approach for abnormal human activities recognition based on ConvLSTM architecture. Sensors **22**(8), 2946 (2022)